

Data-Driven Incipient Fault Detection and Diagnosis for the Running Gear in High-Speed Trains

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Abstract—Incipient fault detection and diagnosis (FDD) is an important measure to improve the efficient, safe and stable operation of high-speed trains. This paper proposes a data-driven FDD method, namely deep slow feature analysis and belief rule base method (DSFA-BRB), for the running gears of high-speed trains. The method uses two kinds of statistics to perform fault detection on the multi-dimensional data of the running gears. In addition, the characteristics of more accurate data are extracted, which greatly reduces the complexity of constructing a diagnostic and quantitative model. Further, by constructing a BRB model combining expert knowledge and data, it is possible to avoid misjudgment caused by data incompleteness. Compared with the traditional methods, the DSFA-BRB algorithm has better performance in reducing fault alarm probability. Finally, the validity of the algorithm is verified by the actual running gears system.

Index Terms—Data-driven fault detection and diagnosis, incipient fault, the running gears, deep slow feature analysis, belief rule base.

I. INTRODUCTION

WITH the continuous improvement of the stability and safety of high-speed trains, research on fault detection and diagnosis of running gears has attracted wide attention. As a key equipment for supporting the car body, transmitting loads, providing braking and traction, the running gears is in high-speed operation for a long time, and it is prone to incipient fault [1]–[3]. However, the incipient fault of the running gear is

covered by various unknown interferences and noises, and it is difficult to effectively diagnose the incipient faults. This paper designs a data-driven fault detection and diagnosis method for the running gear to ensure the safe, stable and efficient operation.

At present, there are many reasons for the incipient faults in the running gears, which can be divided into mechanical faults, electrical faults and sensor faults [4]–[6]. For example, the dynamic interaction between axle box bearings and wheelsets, bogie frame and other components, the excitation caused by gear meshing and track irregularity, the erosion of traction motor drive under high voltage, and the acquisition of noise data by sensors under interference conditions. All of the above can increase the probability of occurrence of incipient faults [7]–[9]. In addition, degradation is also one of the factors that cause incipient faults in the running gears. Due to the inaccurate replacement of components, a potential chain failure between components occurs. Eventually, the overall performance of the running gears system is degraded. Therefore, the incipient faults in the running gears mainly have the following three characteristics:

- 1) Slow: It has the characteristics of small amplitude and slow development. It is difficult to detect the occurrence of faults, but with the passage of time will cause serious consequences, e.g. wear faults of the bearing;
- 2) Sudden: It has the characteristics of small amplitude and fast transients. During the faults of components or subsystems, the local area quickly spreads to the entire system, causing the system to collapse, e.g. short circuit of the traction system;
- 3) Intermittent: It has the characteristics of small amplitude and time-lapse. The gradual degradation of components is the main cause of the occurrence of intermittent incipient faults, e.g. gear components aging.

To solve the incipient faults, existing fault diagnosis methods can be divided into three categories: model-based methods [10]–[12], data-driven methods [13]–[15] and semi-quantitative information methods [16]–[18]. With the increasing coupling degree of modern engineering systems and the uncertainty of various interference environments, it is difficult to establish an accurate analytical model method. Taking the high-speed train running gears as an example, its main components include wheel sets, axle boxes, frames, basic brakes, active suspension devices and motor drives. Among them, the faults of a single main component inside the active suspension system includes air springs, lateral shock absorbers, and anti-snake shock absorbers [1], [19],

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[20]. As a result, data-driven approaches have evolved over the past two decades.

Many scholars have done a large number of work in data-driven fault detection and diagnosis methods [21]–[25]. As pointed out in [7], considering the non-Gaussian characteristics of high-speed trains, a fault detection method based on generalized regular correlation analysis is proposed. It can maximize fault detection capabilities with acceptable false positives. Aiming at the nonlinear problem, a fault detection index called inter-variable variance is presented. And the fault isolation strategy is used to distinguish different types of faults and determine the fault location [8]. Based on the research in [26], the fault is extended to a tensor space, and the convolutional neural network is used to realize automatic learning and classification of fault features, which verifies the accuracy and convergence speed. However, none of the above methods can achieve good results for incipient faults. The main reason is that the correlation algorithm cannot eliminate the interference signal in the weak signal, and cannot detect the changes of incipient fault.

To prevent incipient faults from evolving into permanent faults, incipient faults have become one of the hotspots of current research [13]–[15], [27], [28]. As pointed out in [5], [6], the method based on deep PCA and probability correlation PCA have been developed rapidly. They are applied to the detection of incipient faults in electrical drives, fully excavating the distribution information of incipient faults, whose purposes are successful detection of incipient faults before they develop into permanent faults. Sensor incipient fault detection method based on wavelet transform and kernel PCA is presented. The sensor faults and interference factors leading to errors are analyzed [29]. Although the current improved PCA algorithm provides effectively technical support for solving dynamic process monitoring and incipient fault diagnosis, the probability of false alarm will be greatly increased under time-varying behavior of incipient fault diagnosis. Based on the research in [30], the process monitoring strategy based on slow feature analysis (SFA) is proposed. Two new indicators, S_d^2 and S_e^2 , are used to distinguish abnormal dynamics from dynamic deviation under normal operating conditions. In addition, the recursive adaptive slow feature analysis structure and the partial least squares method are presented to judge the fault degree, and the physical meaning of different monitoring state data is effectively extracted [31], [32]. It is worth noting that the similarities in the above PCA and SFA methods are that there can be a low-dimensional mapping subspace. But the difference is that the PCA expects as many differences as possible in the feature quantity, while the SFA tries to find slowly changing information for analysis.

Therefore, this paper presents a deep slow feature analysis method and establishes a DSFA-based FDD method for the running gears of high-speed trains. The main novelties and contributions of this method are as follows:

- 1) Through multi-source data processing, the weaker fault features can be extracted than traditional SFA. The data features concealed by noise are screened out to facilitate the detection of incipient faults;
- 2) Taking the running gear system as the research object, the theoretical definition of incipient faults is proposed;

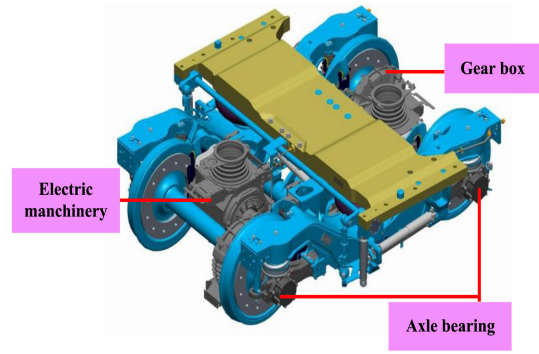


Fig. 1. The structure of running gears.

TABLE I
FAULT DETECTION AND DIAGNOSIS MODEL FOR THE RUNNING GEARS

Test subsystem	Testing model attributes
Bogie	Stability evaluation and prediction model
Bogie	Fault diagnosis model of scratch and peeling
Bogie-wheelset	Wheelset structure parameter prediction model
Bogie-wheelset-bearing	Bearing fault diagnosis and early warning model
Bogie-gearbox-bearing	Gearbox and bearing early warning model
Bogie-motor-bearing	Motor and bearing early warning model

- 3) Combining expert knowledge and data, BRB method is used to ensure the accuracy of fault quantification, and avoid misjudgment caused by data incompleteness;
- 4) It has the sensitivity to improve the incipient fault of the running gears and reduce the false alarm rate of incipient faults.

The remainder of this paper is organized as follows. Section II introduces the running gears system and describes its incipient faults; Section III provides the DSFA-BRB algorithm for the fault detection and diagnosis; Then, based on the experiment of high-speed train running gears, the process of detection and diagnosis is discussed in Section IV. Finally, Section V gives the conclusion drawn from the experimental results and makes the future prospects.

II. PRELIMINARIES

A. The Running Gears of High-Speed Train

Fig. 1 shows the structure of the running gears in high-speed trains. In general, the faults and maintenance of the running gears in practical engineering mainly depends on the collection of various sensor data, including speed, acceleration, temperature, vibration and impact, etc. However, temperature, vibration and impact data are key reference standards for providing faults status in the running gears. The location of the faults point can be determined by observing the trend change of the data.

At present, the classification of the running gears test model is shown in Table I, and the measured component properties are given in detail in the table. In this paper, we study the state change of the running gears, and select the attribute for the bogie-motor-bearing subsystem for analysis. By monitoring the data

information of the local subsystem, the location of the incipient fault is further analyzed.

B. Problem Formulation

We consider the running gears system model: Given the following fault model

$$x = x^* + f + \varepsilon \quad (1)$$

Among them, x^* is the normal system, f denotes the part of incipient faults, ε denotes the interference and noise part. In general, it is difficult to detect and separate incipient faults f from interference and noise ε effectively. f can be caused by many factors and thus

$$f = [f_1, f_2, \dots, f_i] \in \mathbb{R}^{m \times i} \quad (2)$$

According to the existing research and practical engineering requirements of the running gears, its main fault feature data includes: temperature, vibration and impact. From the existing research [6], [20], [21], this paper defines the incipient fault of the running gears as: (1) In terms of qualification: incipient faults do not trigger fault alarms. In other words, in the absence of obvious mechanical faults of gears and electrical faults of motors, the incipient fault data collected by sensors is potentially dangerous for train operation. At the same time, incipient faults can evolve into serious faults at any time, causing the propagation of faults. (2) In terms of quantification, the amplitude of acquisition signal indicates the trigger of incipient fault. The basic deviation of the collected signal level between 1% and 25% is defined as an incipient fault. By using equation (1), the statistical model of incipient faults can be described as

$$\begin{aligned} x &= x^* + f + \varepsilon \\ \text{s.t. } \|f./x\| &\leq 25\% \end{aligned} \quad (3)$$

On the other hand, the running gears diagnosis process requires the construction of a nonlinear model to measure the safety of its state. It can be expressed as

$$y = \mathbf{F}(x^*, f, \varepsilon) \quad (4)$$

Where y is the final results of the diagnosis, and $\mathbf{F}(\bullet)$ is the nonlinear mapping relationship between the data of the running gears and the final results. In this paper, the major tasks are to solve incipient fault detection and the nonlinear model of diagnosis.

III. METHODOLOGY

In this section, the DSFA method is proposed in combination with the conventional SFA algorithm, and is applied in the incipient fault detection. Then, the FDD is given for the incipient fault of running gears.

A. Conventional SFA

The purpose of SFA is to find the mapping relationship between the objective function and the input function, this is $s_j = g_j(x)$, and extract more effective information [30]–[32]. When the transformation function $\{g_j(\bullet), j = 1, 2, \dots, m\}$ is a linear slow feature analysis, the slowness feature $s_j(k)$ can

be represented by the input data $\mathbf{x} = [x_1, x_2, \dots, x_m]^T$. It is evident that

$$s_j = w_j^T x_j \quad (5)$$

Where w_j^T denotes the mapping coefficient matrix of the slow feature, and the objective function and the constraint can be formulated as:

$$\min \Delta(s_j(k)) = \min \langle \dot{s}_j^2 \rangle_k = \min w_j^T \langle \dot{x}_j^T \dot{x}_j \rangle_k w_j \quad (6)$$

$$\langle s_j \rangle_k = w_j^T x_j = 0 \quad (7)$$

$$\langle s_j^2 \rangle_k = w_j^T \langle x_j^T x_j \rangle_k w_j = 1 \quad (8)$$

$$\forall i \neq j, \langle s_i s_j \rangle_k = 0 \quad (9)$$

Where $\langle \cdot \rangle_k$ represents sample averaging. And $\dot{s}_j$ denotes the first order difference of s_j with respect to sampling. The objective function (6) is to extract slow features as much as possible. Constraints (7) and (8) normalize all output signals to a common scale, which makes their temporal derivative directly comparable. Constraint (9) requires that the output signals are decorrelated from one another and guarantees that different output signal components code for different information [33].

The above optimization problem is equivalent to solving the generalized eigenvalue decomposition, and thus

$$\mathbf{A}\mathbf{W} = \mathbf{B}\mathbf{W}\mathbf{\Lambda} \quad (10)$$

Where $\mathbf{\Lambda} = \text{diag}(\lambda_1, \lambda_2, \dots, \lambda_m)$ is the diagonal matrix composed of generalized eigenvalues, $\mathbf{W} = \{w_1, w_2, \dots, w_m\}$ is the corresponding generalized eigenvector matrix, $\mathbf{A} = \langle \dot{\mathbf{x}}^T \dot{\mathbf{x}} \rangle_k$, $\mathbf{B} = \langle \mathbf{x}^T \mathbf{x} \rangle_k$. It is easier to obtain the feature vector $\mathbf{W}$ and get the slow feature $s_j = w_j^T x_j$. Generally, the optimization problem (10) is transformed into two SVD decompositions. The process is as follows:

The first SVD decomposition spheroidizes the data, eliminating the correlation between variables, that is,

$$\mathbf{B} = \langle \mathbf{x}^T \mathbf{x} \rangle = \mathbf{U}\mathbf{\Omega}\mathbf{U}^T \quad (11)$$

After $\mathbf{x}$ is standardized, the spheroidization data can be expressed as

$$\mathbf{z} = \mathbf{\Omega}^{-\frac{1}{2}} \mathbf{U}^T \mathbf{x} = \mathbf{Q}^T \mathbf{x} \quad (12)$$

Where the spheroidization matrix is $\mathbf{Q} = \mathbf{\Omega}^{-\frac{1}{2}} \mathbf{U}^T$, and the spheroidized data retain the above constraints $\langle \mathbf{z} \rangle_k = \mathbf{Q}^T \langle \mathbf{x} \rangle_k = 0$, $\langle \mathbf{z}^T \mathbf{z} \rangle_k = \mathbf{Q} \langle \mathbf{x}^T \mathbf{x} \rangle_k \mathbf{Q}^T = \mathbf{I}$. On this basis, the slow feature expression is $s_j = w_j^T x_j = \mathbf{W}(\mathbf{Q}^T)^{-1} \mathbf{z} = \mathbf{P}\mathbf{z}$. Therefore, solving $\mathbf{W}$ is equivalent to solving matrix $\mathbf{P}$. From the equations (7) and (8), the slow feature s_j satisfies the constraint that the mean value is 0 and the variance is 1, that is,

$$\langle s_j^T s_j \rangle_k = \mathbf{P}^T \langle \mathbf{z}^T \mathbf{z} \rangle_k \mathbf{P} = \mathbf{P}^T \mathbf{P} = \mathbf{I} \quad (13)$$

Obviously, the matrix $\mathbf{P}$ in equation (13) must be an orthogonal matrix. Therefore, the optimization problem of equation (9) is further transformed into solving an orthogonal matrix $\mathbf{P}$, so that $\mathbf{Q}$ is minimal. So the optimization problem of equation

(10) can be solved by SVD for $\langle \dot{\mathbf{z}}^T \dot{\mathbf{z}} \rangle_k$. The second SVD decomposition, $\langle \dot{\mathbf{z}}^T \dot{\mathbf{z}} \rangle_k = \mathbf{P} \mathbf{\Lambda} \mathbf{P}^T$ can be obtained for the $\langle \dot{\mathbf{z}}^T \dot{\mathbf{z}} \rangle_k$ decomposition. Further, a feature vector $\mathbf{W}$ can be obtained, that is,

$$\mathbf{W} = \mathbf{P} \mathbf{Q} = \mathbf{P} \mathbf{\Omega}^{-1/2} \mathbf{U}^T \quad (14)$$

In other words, after the feature vector $\mathbf{W}$ is obtained, the slow feature $\mathbf{s}$ can be expressed as $\mathbf{s} = \mathbf{P} \mathbf{z} = \mathbf{W} \mathbf{x}$.

B. DSFA-Based Incipient-Fault Detection

According to the conventional SFA algorithm, a data-driven method based on DSFA is proposed by using the concept of function decomposition. The slow feature vector is continuously decomposed to achieve the purpose of detecting incipient faults in the running gears of the high-speed train. The specific derivation process will be described below. It is assumed that the normalized data of the running gears is $\mathbf{x} = [x_1, x_2, \dots, x_m] \in \mathbb{R}_{n \times m}$. The selected features and samples respectively are m and n . In the first SVD, different mappings are made in subspace according to $\mathbf{x}$. After spheroidizing the data $\mathbf{z} = [z_1, z_2, \dots, z_m]$, $\mathbf{z}$ still satisfies the constraints (7)–(9). According to the conventional SFA method, $\langle \dot{\mathbf{z}}^2 \rangle$ is minimized, and $\langle \dot{\mathbf{z}}^T \dot{\mathbf{z}} \rangle_k$ is decomposed by SVD, that is,

$$\mathbf{s} = \langle \dot{\mathbf{z}}^T \dot{\mathbf{z}} \rangle_k = \mathbf{P} \mathbf{\Lambda} \mathbf{P}^T \quad (15)$$

The slowness of each feature can be expressed as $\Delta s_j = \frac{w_j^T \mathbf{A} w_j}{w_j^T \mathbf{B} w_j} = \frac{w_j^T \lambda_j \mathbf{B} w_j}{w_j^T \mathbf{B} w_j} = \lambda_j$. With reference to the different slowness Δs_j , the number of principal component slowness l can be determined [30]. Equation (15) is decomposed into $\mathbf{P}$ and $\mathbf{\Lambda}$ to obtain $\mathbf{P} = [\mathbf{P}_d, \mathbf{P}_r]$ ($\mathbf{P}_d \in \mathbb{R}^{m \times l}$, $\mathbf{P}_r \in \mathbb{R}^{(l+1) \times m}$) and $\mathbf{\Lambda}_d = \text{diag}(\sigma_1^2, \sigma_2^2, \dots, \sigma_l^2)$, $\mathbf{\Lambda}_r = \text{diag}(\sigma_{l+1}^2, \sigma_{l+2}^2, \dots, \sigma_m^2)$. Two subspaces of m -dimensional measurement subspace, i.e. principal slowness subspace and residual slowness subspace, are generated

$$\mathbf{s} = [\mathbf{s}_d, \mathbf{s}_e]^T \quad (16)$$

$$\mathbf{s}_d = \mathbf{P}_{1,N}^T \mathbf{\Lambda}^{-1/2} \mathbf{U}^T \mathbf{x} \quad (17)$$

$$\mathbf{s}_e = \mathbf{P}_{(N+1);m}^T \mathbf{\Lambda}^{-1/2} \mathbf{U}^T \mathbf{x} \quad (18)$$

Where N is the number of slow features that account for the main component, which is determined by the criteria defined in [29]. In the data space of input variables, slow feature can be decomposed into principal slow feature $\mathbf{s}_d$ and residual slow feature $\mathbf{s}_e$.

The method description and steps of DSFA is shown in Fig. 2 and Fig. 3. For equation (15), the obtained slow feature can be further decomposed into representations, and thus

$$\mathbf{s} = \left[[\mathbf{s}_d^{n,1}, \mathbf{s}_e^{n,1}]^T, [\mathbf{s}_d^{n,2}, \mathbf{s}_e^{n,2}]^T, \dots, [\mathbf{s}_d^{n,i}, \mathbf{s}_e^{n,i}]^T \right] \quad (19)$$

$$\mathbf{s}_d^{n,t} = \begin{cases} \mathbf{P}_n^T \mathbf{\Lambda}^{-1/2} \mathbf{U}_{n(i;j),t}^T \mathbf{x} & t = \text{odd} \\ \mathbf{P}_n^T \mathbf{\Lambda}^{-1/2} (\mathbf{I} - \mathbf{U}_{n(i;j),t}^T \mathbf{U}_{n(i;j),t}^T)^{-1/2} \mathbf{x} & t = \text{even} \end{cases} \quad (20)$$

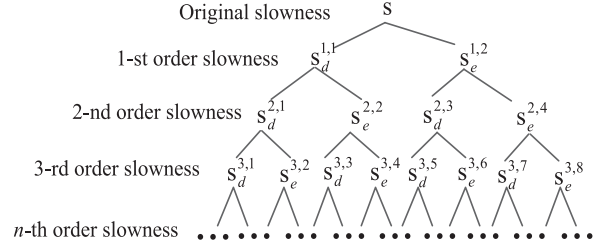


Fig. 2. The structure of DSFA.

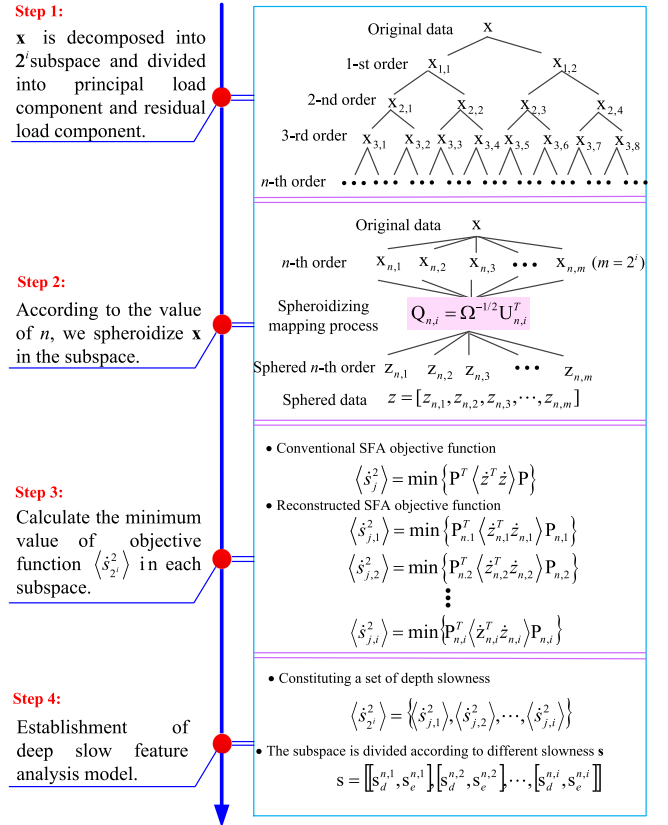


Fig. 3. The steps of DSFA.

$$\mathbf{s}_e^{n,t} = \begin{cases} \mathbf{P}_n^T \mathbf{\Lambda}^{-1/2} \mathbf{U}_{n(j+1;i),t}^T \mathbf{x} & t = \text{odd} \\ \mathbf{P}_n^T \mathbf{\Lambda}^{-1/2} (\mathbf{I} - \mathbf{U}_{n(j+1;i),t}^T \mathbf{U}_{n(j+1;i),t}^T)^{-1/2} \mathbf{x} & t = \text{even} \end{cases} \quad (21)$$

where n is the order of slowness decomposition, i is the number of slowness s decomposition.

The original data set $\mathbf{x}$ of the high-speed train running gears can be multi-mapped by equations (18) and (19) to improve the ability to monitor incipient faults. The DSFA method can be used to analyze the status of the running gears under complex conditions, so it is important to establish reasonable monitoring indicators for the running gears. Designing Hotelling's T^2 test statistic is applied to $\mathbf{s}_d$ and $\mathbf{s}_e$ to monitor system variation [30]. Hotelling's T^2 test statistics are defined by

$$T^2 = \mathbf{s}_d^T \mathbf{s}_d \quad (22)$$

$$T_e^2 = \mathbf{s}_e^T \mathbf{s}_e \quad (23)$$

Besides, the slow variation $\dot{s} = [\dot{s}_1, \dot{s}_2, \dots, \dot{s}_m]^T = [\dot{s}_d^T, \dot{s}_e^T]^T$ can also monitor the slowness feature and further describe the state variation of the running gears by defining the S^2 and S_e^2 statistics [30]. These two statistics are calculated as

$$S^2 = \dot{s}_d^T \Lambda_d^{-1} \dot{s}_d \quad (24)$$

$$S_e^2 = \dot{s}_e^T \Lambda_e^{-1} \dot{s}_e \quad (25)$$

Therefore, the Hotelling's T^2 and the S^2 test statistic of the DSFA are defined by

$$T_{j,t}^2 = (\mathbf{s}_d^{n,t})^T (\mathbf{s}_d^{n,t}) \quad (26)$$

$$S_{j,t}^2 = (\dot{\mathbf{s}}_d^{n,t})^T \Lambda_d^{-1} (\dot{\mathbf{s}}_d^{n,t}) \quad (27)$$

where $T_{j,t}^2$ and $S_{j,t}^2$ are represented as the principal slowness feature, which can be used as the statistic to detect the running gears faults.

The boundaries of $T_{j,t}^2$ and $S_{j,t}^2$ will fluctuate within a specific local range. Therefore, the control limits J of $T_{j,t}^2$ and $S_{j,t}^2$ can be determined by the probability distribution [30]–[32]. In order to use the above indicators for detection, the control limits should be estimated by using data, and the calculation J can be referred to [30]. The detection strategy is summarized as follows:

- 1) If $T_{j,t}^2 \leq J_{T^2}$ and $S_{j,t}^2 \leq J_{S^2}$, the running gears is in normal operation;
- 2) If $T_{j,t}^2 \geq J_{T^2}$ and $S_{j,t}^2 \leq J_{S^2}$, the running gears is in normal operation, but it can be regarded as caused by the variable behavior of the system. For example, man-made condition switching, complex environment mutation and interference;
- 3) If $S_{j,t}^2 \geq J_{S^2}$, the status of the running gears is a incipient fault.

In summary, the monitoring data variation do not satisfy the statistical regularity, and the interference is completely random in the running department system. The DSFA method is used to detect the state of the running gears, and different test statistics also detect incipient faults and variable behavior.

C. Diagnosis of Incipient Faults

Fault diagnosis is a critical issue when the DSFA method successfully detects an initial fault. As a new semi-quantitative analysis method, BRB algorithm can effectively solve the fault diagnosis of small sample data, especially the nonlinear and non-Gaussian system models can be analyzed with data and expert experience. Description of BRB fault diagnosis method for the running gears is given

R_k : If x_1 is $A_1^k \wedge x_2$ is $A_2^k \cdots \wedge x_M$ is A_M^k

Then $\{(D_1, \beta_{1,k}), \dots, (D_N, \beta_{N,k})\}$

With a rule weight θ_k

And attribute weight $\delta_1, \delta_2, \dots, \delta_M$ (28)

Where $x_i (i = 1, 2, \dots, M)$ is the i -th antecedent attribute for the state of running gears; $A_i^k (i = 1, 2, \dots, M; k = 1, 2, \dots, L)$ is the value of the x_i -th antecedent attribute in the k -th rule; L is the total number of rules; $\beta_{j,k}$ is the belief degree

of conclusion in the k -th rule for the j -th diagnostic result D_j ; θ_k is the rule weight of k -th rule; and $\delta_{i,k}$ is the weight of the i -th antecedent attribute.

Consider the different dimension issues of the running gears feature data. Calculate the antecedent attribute matching degree to better describe the degree of matching between the antecedent attribute and the rule, that is,

$$\varphi_{i,j}^k = \begin{cases} 1 - \varphi_{i,j}^k, & k = l + 1; \\ \frac{D_{i+1}^{l+1} - x_i}{D_{i+1}^{l+1} - D_i^l}, & k = l (D_i^l \leq x_i \leq D_{i+1}^{l+1}); \\ 0, & k = 1, 2, \dots, N (k \neq l, l + 1). \end{cases} \quad (29)$$

There is a different interaction between the belief rules formed by the expert knowledge of the running gears and the matching degree of the antecedent attributes. The weighted w_k is activated by calculating the k -th belief rule, and some of the belief rules in the BRB are activated, thereby performing reasonable weight division on different matching degrees. Therefore, the activation degree w_k of the k -th rule can be expressed as

$$w_k = \frac{\theta_k \prod_{i=1}^M (\varphi_i^k)^{\bar{\delta}_i}}{\sum_{i=1}^L \alpha_i \prod_{i=1}^M (\varphi_i^i)^{\bar{\delta}_i}} \quad (30)$$

$$\bar{\delta}_i = \frac{\delta_i}{\max_{i=1,2,\dots,M} \{\delta_i\}} \quad (31)$$

Where $w_k \in [0, 1]$, $k = 1, 2, \dots, L$, $\bar{\delta}_i$ denote attribute weights.

Evidence reasoning algorithm can get the fault diagnosis BRB output $S(x)$ of the running gears [17], [18], that is,

$$S(x) = (Y_j, \hat{\beta}_j), j = 1, 2, \dots, N \quad (32)$$

Where $\hat{\beta}_j$ is relative to the belief degree of diagnostic results Y_j . Thus, we have that

$$\begin{aligned} \hat{\beta}_j &= \frac{\mu \times [\prod_{k=l}^L (w_k \beta_{j,k} + \rho) - \prod_{k=l}^L \rho]}{1 - \mu \times [\prod_{k=1}^L (1 - w_k)]} \\ \mu &= \left[\sum_{j=1}^N \prod_{k=l}^L (w_k \beta_{j,k} + \rho) - (N - 1) \prod_{k=1}^L \rho \right]^{-1} \\ \rho &= 1 - w_k \sum_{i=1}^N \beta_{i,k} \end{aligned} \quad (33)$$

Where w_k can be calculated using equation (30); $\hat{\beta}_j$ is a function of the rule weight θ_k ($k = 1, 2, \dots, L$), the belief degree $\beta_{j,k}$ ($j = 1, 2, \dots, N; k = 1, 2, \dots, L$) and $\bar{\delta}_i$ ($i = 1, 2, \dots, L$) is the attribute weight.

The expected utility of the final running gears $S(x)$ can be obtained by

$$y = \mu(S(x)) = \sum_{j=1}^N \mu(Y_j) \beta(j) \quad (34)$$

The expectant utility y can reflect the fault quantification status of running gears.

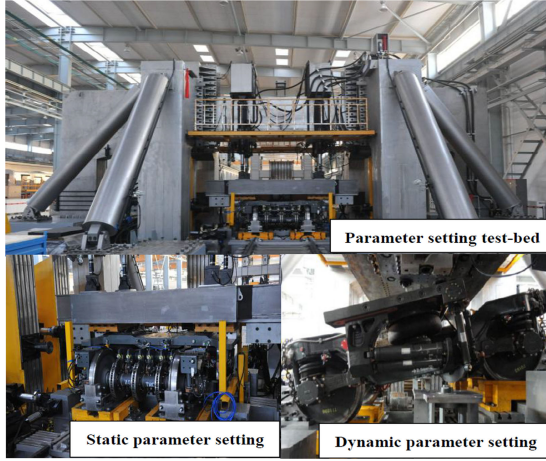


Fig. 4. The running gears posture parameters operating platform.

D. Incipient Fault Detection and Diagnosis Strategies of Running Gears

Through the above two parts B and C, the incipient FDD of running gears can be divided into the following steps:

- Step 1:* Obtain the data x from the sensors of the temperature, vibration and impact features of the running gears;
- Step 2:* Spheroidizing the data x according to equation (12) to obtain spheroidized data z ;
- Step 3:* For the first-order difference of the spheroidized data z , the minimum of the objective function is obtained by the equation (15);
- Step 4:* According to the running gears under different noise conditions, the slowness s is decomposed to different degrees by equation (19);
- Step 5:* By designing the new statistic $T_{j,k}^2$ and $S_{j,k}^2$, the state of running gears is detected and a new physical significance is given;
- Step 6:* According to expert knowledge, the attribute weight, reference value and initial belief are given. And the parameters of equation (28) belief rule base are determined;
- Step 7:* Establish a belief rule base fault diagnosis model by using equations (29)–(33). The state of the running gear is estimated according to the equation (34).

Remark 1: In practical engineering, the fault running gears generally finds changes in the collected data, and then the professional person checks the inspection site, a large amount of manpower cost is consumed. In addition, the incipient faults about the running gears are difficult to detect only by observing data changes. Therefore, this paper uses the new test statistic $T_{j,k}^2$ and $S_{j,k}^2$ to detect and analyze the incipient faults of the running gears, and the fault diagnosis is judged by belief rule base.

IV. EXPERIMENTAL RESULTS AND ILLUSTRATION

The dynamic and static parameters test-bed for simulating the movement attitude of high-speed train running gears shown in Fig. 4 is developed by CRRC Changchun Railway Vehicles



Fig. 5. Simulated running test bench.

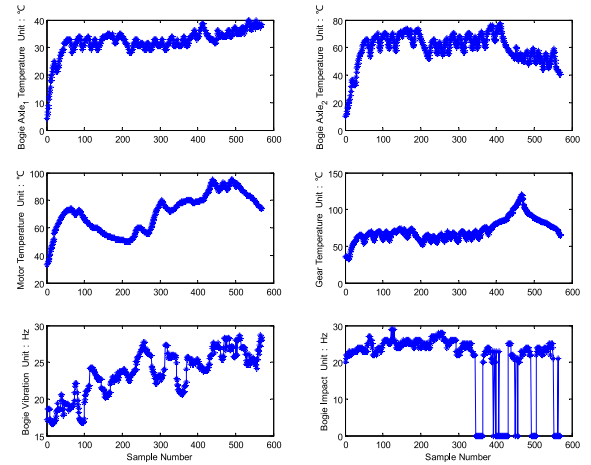


Fig. 6. Data collected by six sensors.

Co., Ltd. Taking the actual operation train as an example, by combining the simulated operation test-bed in Fig. 5 with the actual train operation data and the fault record table, the incipient fault information before the fault occurs is intercepted. According to the situation that the speed of the train engine is over 1000 r/min and tends to run steadily, the method is validated by collecting signals from six sensors.

According to the historical data, 570 samples are collected from each sensor. When 200 samples are taken, an incipient fault as defined in equation (3) is injected into the motor. When 400 samples are taken, the temperature of the motor increases gradually, and the data collected by the gearbox and the bogie also change obviously because of the coupling between the components, as shown in Fig. 6. Therefore, the validation data is $x = \{x_1, x_2, x_3, x_4, x_5, x_6\} \in \mathbb{R}^{570 \times 6}$, in which we intercept 300 samples for training and 270 samples for testing. And the test data is injected after 150 samples to ensure the reliability of the proposed algorithm.

Remark 2: The sensor of the monitors running gear collects data every minute. The data in Fig. 6 is obtained by taking an average value of the original data every four points. Moreover, the temperature sensor is in close contact with the bearing, gear

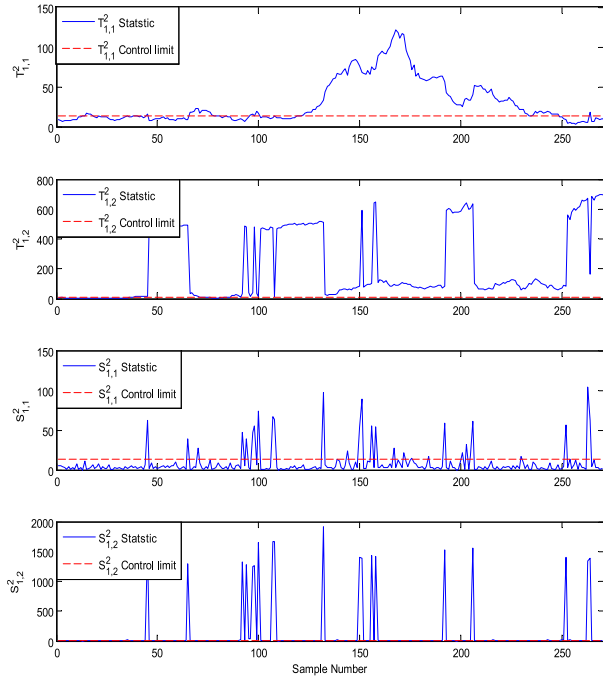
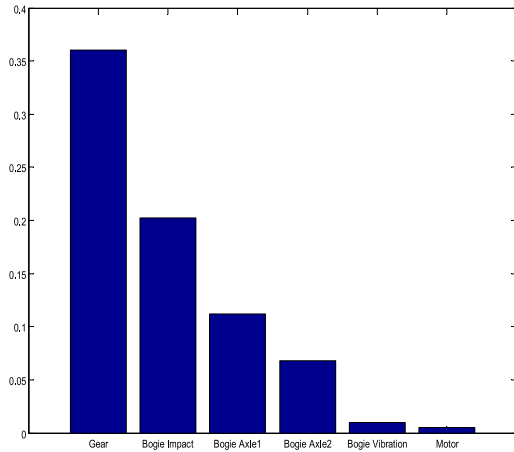
Fig. 7. Fault detection results for f with the 1-st order DSFA.

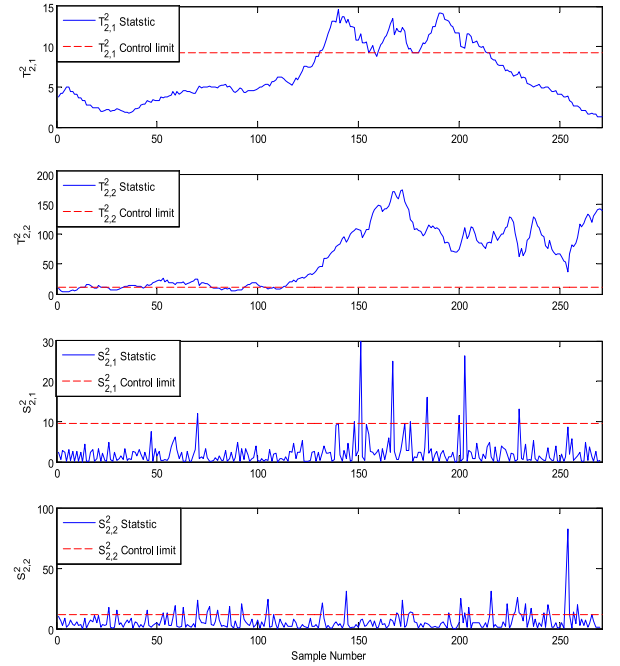
Fig. 8. Slowness value of different feature with the 1-st order DSFA.

box and motor components, so that the external interference noise is relatively small. Compared to vibration and impact sensors, random noise is brought by the influence of train speed and acceleration.

Remark 3: There are two reasons why bogie impact data is zero: on the one hand, the result of fault injection. On the other hand, the impact data often fluctuates in the zero range. Therefore, the method of median filtering is used to pre-process the original data.

A. Incipient Fault Detection

Based on the analysis in Section III, The 1-order DSFA is the standard SFA, and the test statistics $T_{1,1}^2$ and $S_{1,1}^2$ can be given by equations (26) and (27), i.e., $j = 1, t = 1$. Fig. 7 shows the results of the 1st-order slow feature fault detection, and Fig. 8

Fig. 9. Fault detection results for f with the 2-nd order DSFA.

shows the slowness values of the different features. We find that the gearbox temperature is not the slowest. According to the slowness reduction principle [30], the principal component number is 5, and the residual component number is 1, and the mapping coefficient matrix $\mathbf{W}_1$ of the slow feature is obtained. Based on the monitoring strategy, the 1st-order DSFA of running gears detection results is in a fault state, but the alarm will always be heard due to external interference and noise.

$$\mathbf{W}_1 = \begin{bmatrix} -1.34 & 2.71 & 1.25 & -0.13 & 0.35 & -0.05 \\ 1.33 & -3.16 & 0.28 & -0.41 & 0.44 & 0.17 \\ -0.74 & 0.04 & -0.48 & 1.12 & 0.19 & -0.01 \\ 1.27 & 0.30 & -0.03 & 0.13 & 0.02 & -0.01 \\ 0.01 & 0.04 & -0.06 & 0.75 & -0.51 & 0.91 \\ -0.35 & 0.34 & -1.43 & -0.56 & 0.19 & 0.06 \end{bmatrix} \quad (35)$$

As a result, Fig. 9 shows that we further in-depth analysis to obtain the 2-nd order DSFA result in the principal component number. According to the fault detection strategy proposed in this paper, the test results before sampling 135 are $T_{2,1}^2 < 0$ and $S_{2,1}^2 < 0$ are normal condition; the results between samples 135-150 are $T_{2,1}^2 > 0$ and $S_{2,1}^2 < 0$ are normal deviations; the result after sampling 150 is that $S_{2,1}^2 > 0$ frequent triggers are incipient faults. It can be clearly seen that the $T_{2,1}^2$ and $S_{2,1}^2$ test statistics detect the fault of the running gears. And different slowness values are given in Fig. 10. At the same time, the mapping coefficient matrix $\mathbf{W}_2$ of the slow feature is obtained, and the related number of principal component is 3 and residual component is 2. Based on the principle of minimum slowness, the motor temperature and bogie vibration feature can be

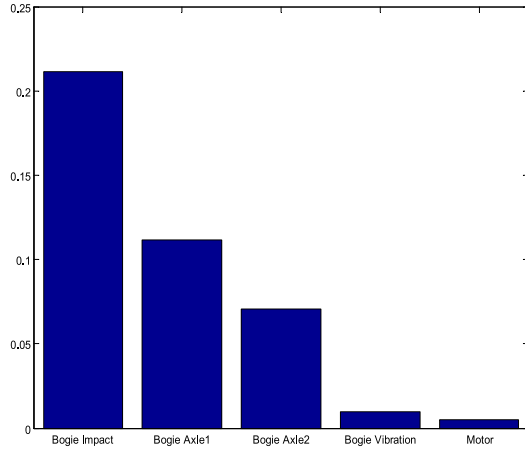


Fig. 10. Slowness value of different feature with the 2-nd order DSFA.

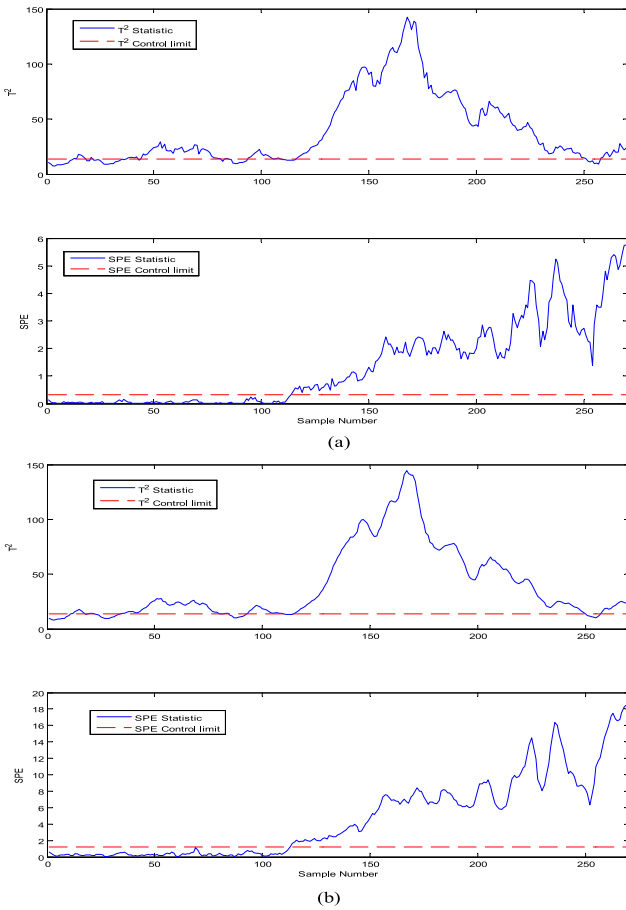


Fig. 11. Monitoring results of (a) PCA and (b) DPCA in running gears.

obtained.

$$W_2 = \begin{bmatrix} 2.95 & 1.26 & 0.13 & -0.36 & 0.05 \\ -3.40 & 0.27 & 0.38 & -0.45 & -0.17 \\ 0.25 & -0.51 & -1.18 & -0.20 & 0.01 \\ 0.06 & -0.06 & -0.73 & 0.52 & -0.91 \\ 0.42 & -1.42 & 0.56 & -0.20 & -0.06 \end{bmatrix} \quad (36)$$

Besides, the comparison of other methods is given in Table II. It is found that the PCA, DPCA and SFA methods

TABLE II
PERFORMANCE COMPARISONS FOR FOUR TYPES OF INCIPIENT FAULTS

Method	PCA	DPCA	SFA	the proposed DSFA
M_d	4	4	5	3
M_e	2	2	1	2
FAR	0.8266	0.8662	0.1111	0.0444

can achieve good detection results, but the DSFA false alarm probability is significantly lower than the other methods, which improves the detection sensitivity of the running gears. The comparison method has different significance. These traditional FDD algorithm such as PCA and DPCA mainly involve the static characteristics of signals. For changes in the system state (normal deviation), SFA and DSFA can analyze both static and dynamic characteristics of the running gears system in parallel.

Remark 4: In Table II, M_d and M_e represent the principal component and the residual component, respectively. And the correlation calculation can be referred to in [30]. The first-order SFA process divides the component components into principal 5 and residual 1, where the residual feature quantity for the gear box temperature.

Remark 5: After the mechanism analysis, the gear box is the intermediate component of the rotation process of the motor and the bearing, and the collected data has certain redundancy. Therefore, in the second-order SFA process, the gearbox data is not analyzed, and yet the principal component data is subjected to slow feature analysis.

B. Incipient Fault Diagnosis

After the DSFA detects a incipient fault, it analyzes the data that mainly causes the fault in the running gears. Meanwhile, a large amount of data is reduced in dimension to obtain motor temperature and bogie vibration. In addition, if only a single fault diagnosis statistic is used for fault diagnosis, empirical analysis of the data lacking the fault history table will cause deviations. Therefore, the BRB method is combined with expert knowledge and fault data to improve the results of the running gears fault diagnosis. The fault level of the running gears can be divided into the following four types:

Normal status: The condition of the running gears is healthy. Temperature and vibration are normal, and the amplitudes are 52–70 °C and 17–20 Hz;

Sub-health status: The running gears does not need timely maintenance, and the high-speed train can be repaired at night. In this case, the temperature value is 71–80 °C, and the vibration value is 21–23 Hz;

Under-fault status: The running gears performs maintenance when the high-speed train is inbound. In this case, the vibration value is 24–26 Hz, and the temperature value is 81–85 °C;

Fault status: The running gears needs to stop and repair immediately. In this case, the vibration is severe, the amplitude is up to 33 Hz, and the temperature is up to 120 °C.

The bogie vibration and motor temperature indicators are important for the fault diagnosis of running gears system, and are essential to the rule weight of the indicator, so w_i can be assigned a value of 1. In order to better quantify the degree of indicators,

TABLE III
TEMPERATURE AND VIBRATION REFERENCE VALUES

Semantic values	H	U	F	N
Temperature quantized vaules	1	2	3	4
Vibration quantized vaules	1	2	3	4

TABLE IV
FAULTS LEVEL REFERENCE VALUES

Semantic values	S ₁	S ₂	S ₃	S ₄
Faults level quantized vaules	1	2	3	4

TABLE V
BELIEF RULE BASE PARAMETER SETTING

Rule number	Vibration and temperature	Faults level $\{S_1, S_2, S_3, S_4\} = \{1, 2, 3, 4\}$
1	H and H	$\{(S_1, 0.9), (S_2, 0.1), (S_3, 0), (S_4, 0)\}$
2	H and U	$\{(S_1, 0.5), (S_2, 0.5), (S_3, 0), (S_4, 0)\}$
3	H and F	$\{(S_1, 0.2), (S_2, 0.5), (S_3, 0.3), (S_4, 0)\}$
4	H and N	$\{(S_1, 0.1), (S_2, 0.2), (S_3, 0.3), (S_4, 0.5)\}$
5	U and H	$\{(S_1, 0.6), (S_2, 0.4), (S_3, 0), (S_4, 0)\}$
6	U and U	$\{(S_1, 0.2), (S_2, 0.6), (S_3, 0.2), (S_4, 0)\}$
7	U and F	$\{(S_1, 0), (S_2, 0.5), (S_3, 0.4), (S_4, 0.1)\}$
8	U and N	$\{(S_1, 0), (S_2, 0.1), (S_3, 0.2), (S_4, 0.7)\}$
9	F and H	$\{(S_1, 0.2), (S_2, 0.3), (S_3, 0.5), (S_4, 0)\}$
10	F and U	$\{(S_1, 0), (S_2, 0.5), (S_3, 0.5), (S_4, 0)\}$
11	F and F	$\{(S_1, 0), (S_2, 0.1), (S_3, 0.8), (S_4, 0.1)\}$
12	F and N	$\{(S_1, 0), (S_2, 0), (S_3, 0.2), (S_4, 0.8)\}$
13	N and H	$\{(S_1, 0.1), (S_2, 0.1), (S_3, 0.2), (S_4, 0.6)\}$
14	N and U	$\{(S_1, 0), (S_2, 0.1), (S_3, 0.2), (S_4, 0.7)\}$
15	N and F	$\{(S_1, 0), (S_2, 0), (S_3, 0.2), (S_4, 0.8)\}$
16	N and N	$\{(S_1, 0), (S_2, 0), (S_3, 0), (S_4, 0.1)\}$

there are four reference levels for temperature, vibration, and fault levels in Tables III and IV, where bogie vibration and motor temperature semantics are normal, and general, medium, and poor are simply recorded as H, U, F, N, and fault level semantics in normal, sub-health, under-fault and fault states, simply referred to as S₁, S₂, S₃ and S₄. Since the bogie vibration and motor temperature reference values are divided into four levels, there are a total of 16 belief rules. Based on the expert knowledge, the set BRB parameters are given in Table V, and the fault diagnosis model of the running gear is established. The k -th rule can be described as

$$R_k : \text{If Temperature is } A_1^k \wedge \text{Vibration is } A_2^k \\ \text{Then fault level is} \\ \{(1, \beta_{1,k}), (2, \beta_{2,k}), (3, \beta_{3,k}), (4, \beta_{4,k})\} \quad (37)$$

Remark 6: In the first rule, H AND H means that the bogie vibration and motor temperature is in a normal state. Therefore, the fault level of the running gears is the normal state, and the belief value of the diagnosis is (S₁, 0.9), (S₂, 0.1), (S₃, 0), (S₄, 0).

Fig. 12 shows the BRB diagnostic model for the running gears, and Fig. 13 shows the bogie vibration and motor temperature distribution fault levels. The results show that the BRB model can accurately fit the fault level and reflect the state of the running gears system more realistically. Moreover, the sub-health and the under-fault state are difficult to distinguish in actual engineering. It is normal for the diagnostic quantification results to fluctuate under these two conditions.

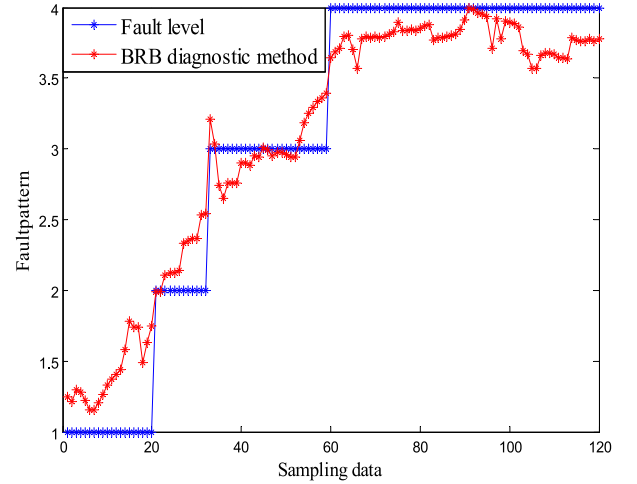


Fig. 12. The BRB diagnostic model for the running gears.

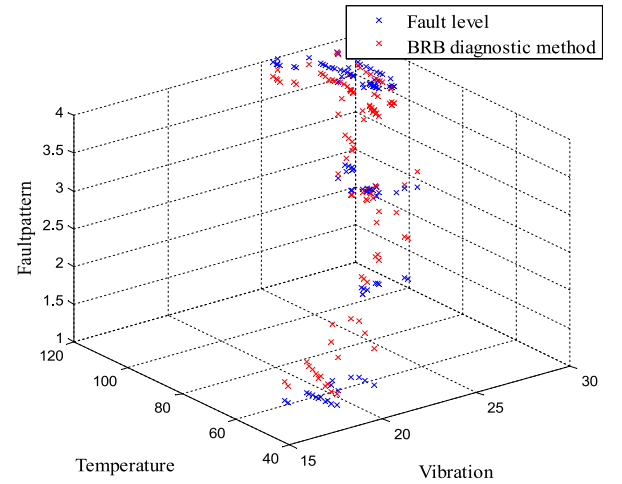


Fig. 13. Data distribution fault levels.

V. CONCLUSION

This paper proposes a data-driven FDD method for the high-speed train running gears. This method deeply analyses the running change and noise information of the running gears, and uses two kinds of test statistics to detect the incipient faults of the running gears in parallel, which overcomes the shortcomings of the traditional multivariate statistical method relying on a single test statistics. Meanwhile, in order to make up for the lack of fault data, the accurate running gears state is given by combining expert knowledge and data. Through the high-speed train running gears test, the excellent performance of the proposed FDD is evaluated. It should be emphasized that: 1) the expert knowledge of the running gears is given by reference to the maintenance guide; 2) the definition of the incipient fault of running gears is not given in the actual project, but is defined by the trend change of the data. Moreover, it is planned to improve the slow feature analysis method and apply it to the general industrial systems.

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